

KUTAISI, Georgia

WMO: 373950

Lat: 42.254N

Lon: 42.628E

Elev: 113

StdP: 99.97

Time Zone: 4.00 (E04)

Period: 97-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-1.4	-0.1	-8.4	1.9	3.8	-6.5	2.2	4.5	21.9	7.2	19.6	9.3	4.0	90	0.468

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	8.9	35.2	23.4	33.0	22.9	31.3	22.6	25.5	31.1	24.8	29.9	24.0	28.8	8.2	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
24.1	19.2	28.1	23.3	18.4	27.5	22.6	17.6	26.8	79.3	31.2	76.1	30.0	73.0	28.9	28.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4)	19.7	16.2	14.2	DB	-3.1	38.4	2.0	1.7	-4.5	39.6	-5.7	40.6	-6.8	41.6	-8.3	42.8
(5)				WB	-4.5	26.9	2.1	0.8	-6.0	27.5	-7.3	28.0	-8.5	28.5	-10.0	29.1

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	16.2	6.9	8.0	10.3	14.4	19.3	23.1	24.8	26.0	22.3	17.8	12.4	8.7	(6)
(7)		DBStd	7.65	3.69	4.36	4.59	4.36	3.96	2.97	2.94	2.87	3.67	3.62	4.10	4.34	(7)
(8)		HDD10.0	352	108	82	53	8	0	0	0	0	0	1	23	77	(8)
(9)		HDD18.3	1607	353	289	250	133	36	2	0	9	55	179	300	(9)	
(10)		CDD10.0	2618	13	27	64	139	289	392	458	494	369	241	96	36	(10)
(11)		CDD18.3	832	0	0	2	15	66	144	200	236	129	37	2	1	(11)
(12)		CDH23.3	6610	0	0	9	119	551	1083	1575	2158	937	170	8	0	(12)
(13)		CDH26.7	2437	0	0	1	18	179	379	622	901	313	24	0	0	(13)
(14)	Wind	WSAvg	4.7	5.1	6.2	5.4	5.0	4.3	4.0	3.4	3.6	4.1	4.6	5.3	5.2	(14)
(15)	Precipitation	PrecAvg	1543	132	120	134	116	105	144	115	103	140	148	143	143	(15)
(16)		PrecMax	1905	318	249	283	170	188	274	227	210	257	344	283	314	(16)
(17)		PrecMin	1210	37	59	61	51	53	79	60	31	22	44	24	35	(17)
(18)		PrecStd	165	65	45	49	29	41	46	44	41	60	71	76	64	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	17.4	20.8	24.2	28.7	33.0	35.1	37.6	38.1	35.0	29.7	24.6	21.2	(19)
(20)		0.4%	MCWB	9.4	12.5	14.2	18.4	20.7	22.5	24.2	23.8	22.9	19.2	15.1	12.3	(20)
(21)		2%	DB	15.4	18.2	21.8	26.4	30.8	32.4	35.0	35.4	32.1	27.2	22.0	18.6	(21)
(22)		2%	MCWB	8.4	10.6	12.9	17.0	19.8	22.4	23.8	23.5	21.9	18.3	13.8	11.1	(22)
(23)		5%	DB	14.0	16.2	19.5	24.5	28.6	30.5	32.1	33.0	29.8	25.4	19.8	16.3	(23)
(24)		5%	MCWB	7.8	9.6	12.0	15.9	19.2	21.8	23.2	23.6	20.9	17.5	13.0	9.8	(24)
(25)		10%	DB	12.4	14.2	17.5	22.2	26.4	28.7	30.0	31.2	28.0	23.6	17.9	14.5	(25)
(26)		10%	MCWB	6.9	8.3	11.0	14.6	18.4	21.3	23.1	23.3	20.3	16.8	12.0	8.9	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	10.1	13.2	15.2	19.4	21.7	25.4	26.5	26.6	24.8	20.2	15.7	13.8	(27)
(28)		0.4%	MCDWB	14.5	19.1	22.7	27.3	29.9	30.8	32.3	32.3	31.3	26.3	22.1	18.1	(28)
(29)		2%	WB	9.2	11.1	13.6	17.6	20.7	24.0	25.4	25.7	23.4	19.2	14.7	11.8	(29)
(30)		2%	MCDWB	13.2	16.6	20.4	25.6	28.5	29.6	31.1	31.1	28.8	24.8	19.7	16.2	(30)
(31)		5%	WB	8.4	10.0	12.4	16.2	19.8	23.0	24.6	25.0	22.3	18.4	13.8	10.6	(31)
(32)		5%	MCDWB	12.4	15.1	18.6	23.5	27.1	28.4	29.8	29.9	27.5	23.8	18.3	15.0	(32)
(33)		10%	WB	7.6	8.9	11.3	15.0	18.8	22.1	23.8	24.3	21.4	17.6	12.7	9.6	(33)
(34)		10%	MCDWB	11.5	13.4	16.5	21.5	25.3	27.1	28.3	28.9	26.2	22.4	16.9	13.8	(34)
(35)	Mean Daily Temperature Range		MDBR	5.9	6.7	7.5	9.1	9.1	8.5	7.7	8.9	8.8	7.8	6.7	6.0	(35)
(36)		5% DB	MCDBR	7.9	10.0	10.9	13.1	12.9	11.9	11.7	12.0	11.5	10.6	9.8	8.4	(36)
(37)		5% DB	MCWBR	4.5	5.7	5.9	6.6	5.9	5.0	4.6	4.8	5.0	5.3	5.5	4.8	(37)
(38)		5% WB	MCDBR	6.9	9.6	10.2	12.5	11.8	10.5	9.7	10.5	10.5	9.4	8.7	7.3	(38)
(39)		5% WB	MCWBR	4.2	5.5	5.7	6.5	5.9	4.8	4.3	4.6	4.8	5.1	5.1	4.4	(39)
(40)		Clear-Sky Solar Irradiance	taub	0.329	0.362	0.415	0.465	0.435	0.424	0.451	0.454	0.401	0.390	0.346	0.323	(40)
(41)	taud		2.303	2.195	2.110	2.057	2.237	2.330	2.279	2.268	2.397	2.373	2.435	2.372	(41)	
(42)	Ebn at Noon		797	823	819	806	844	853	824	808	831	790	780	773	(42)	
(43)	Edh at Noon		87	114	141	160	137	125	131	128	105	96	77	76	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1.63	2.47	3.29	4.46	5.30	5.87	5.80	5.44	4.38	2.96	1.99	1.44	(44)	
(45)		RadStd	0.23	0.30	0.30	0.53	0.54	0.45	0.54	0.41	0.42	0.33	0.26	0.15	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (14 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page